

BATTERY INPUT + PROTECTION

5V CONTROL BUCK (ESP32 + LOGIC ONLY)

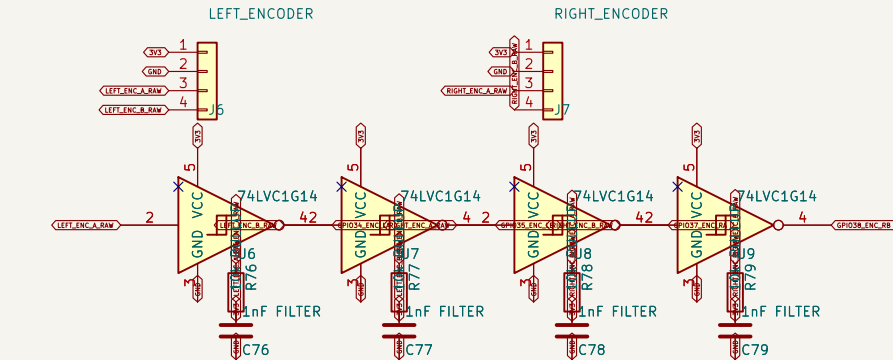
WHEEL ENCODERS (3.3V)

POWER PASS-THROUGH (SEPARATE 5V/5A REGULATOR)

ESP32-PICO-KIT V4.1 SOCKETS

RASPBERRY PI UART / CONTROL

SAFETY / REVIEW NOTES



Encoder inputs include pull-ups, RC filtering, and Schmitt buffers.

- VNH5019 exposed pads need reflow, thermal vias, and large copper pours.
- Verify actual motor stall current before ordering production PCB.
- Pi power is NOT made by U1. Feed J8 from a separate regulated 5V/5A supply.
- Never power ESP32 from USB and EXT_5V simultaneously.
- External fuse and latching E-stop are mandatory.

3-DOF ARM SERVO CONTROL

LEFT MOTOR DRIVER

RIGHT MOTOR DRIVER

Pi power is a separate 5 V / 5 A pass-through branch
DRAFT: verify motor stall current and footprints before manufacture
3S LiPo; 2x GA25-370; provisional 5 A stall per motor

Harrison

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File: pickleball_robot_controller.kicad_sch

Title: Pickleball Robot Wheel Controller + Raspberry Pi Interface

Size: A3 Date: 2026-07-12

Rev: R1-DRAFT

KiCad E.D.A. 10.0.4

Id: 1/1